

```

17. 3) If n is 1, stop. Otherwise, repeat.
18.
19. It is generally thought, but so far not mathematically proven, that
20. every starting number eventually terminates at 1.
21. '''
22.
23. print('Enter a starting number (greater than 0) or QUIT:')
24. response = input('> ')
25.
26. if not response.isdecimal() or response == '0':
27.     print('You must enter an integer greater than 0.')
28.     sys.exit()
29.
30. n = int(response)
31. print(n, end='', flush=True)
32. while n != 1:
33.     if n % 2 == 0: # If n is even...
34.         n = n // 2
35.     else: # Otherwise, n is odd...
36.         n = 3 * n + 1
37.
38.     print(', ' + str(n), end='', flush=True)
39.     time.sleep(0.1)
40. print()

```

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. How many numbers are in a Collatz sequence that begins with 32?
2. How many numbers are in a Collatz sequence that begins with 33?
3. Are the Collatz sequences for starting numbers that are powers of two (2, 4, 8, 16, 32, 64, 128, on so on) always composed of only even numbers (aside from the final 1)?
4. What happens when you enter 0 for the starting integer?